

Quantum-Tube 'Prime Resonator' Presentation

1. Concept Overview

Create a planar array of thin waveguides ('tubes') that implement arithmetic translations T_n as unitary propagation-and-coupling. Each prime p gets a dedicated guide section of physical length $L_p = \alpha \cdot \log p$, with couplers routing amplitude from node n to node $n + p$. The one-step propagator reproduces $H\delta(s) = \sum_n (p^{-s} T_n + p^{-(1-s)} T_n^*)$, with weights realized by phase shifters and tap couplers.

2. System Architecture (Photonic SiN PIC)

Substrate: 200 mm Si wafer with 4–6 μm thermal SiO_2 bottom cladding.

Core: LPCVD SiN, 300 nm thickness (single-mode at 1550 nm with 1.2 μm width).

Top cladding: PECVD SiO_2 , 2–3 μm .

Prime waveguide bank $\{W_p\}$: straight sections with lengths $L_p = \alpha \cdot \log p$.

Coupler matrix: low-loss couplers for forward T and reverse T^* channels.

Phase/attenuation weights: thermo-optic phase shifters impose ϕ and gains.

Node lattice: finite set $N = \{1, \dots, N\}$ with routing from node n to p_n .

Scaling constant $\alpha \approx 4.95 \text{ mm}$ for $\tau = 20 \text{ ps}$ and $n_{\text{eff}} \approx 1.75$.

3. Concrete Device Spec (Rev-A Demo)

Die size: 35 mm × 25 mm (SiN PIC).

Prime set: $p \in \{2,3,5,7,11,13,17,19,23,29\}$.

Node count: $N = 16$.

Waveguide geometry: 300 nm × 1.2 μm; $n_{\text{eff}} \approx 1.75$.

Propagation loss: ≤ 0.2 dB/cm.

Couplers: 3 dB MMI + variable directional couplers with thermo-optic tuners.

Phase shifters: 200 μm heaters; 2π phase at ≈ 25 mW.

I/O: 8 grating couplers for inputs; 16 for outputs.

4. Measurement Protocol (ζ -line Mapping)

Calibration: $\sigma = 1/2$, zero phases ($t=0$), equalize path losses.

Critical-line sweep: fix $\sigma = 1/2$, ramp global phase K to sweep t .

Record $H(t)$: reference input port and composite output.

Zero signatures: deep minima in $|H(t)|$ align with ζ -zeros.

Symmetry check: swap forward/reverse banks to confirm mirrored response.

5. Error/Tolerance Budget

Length error $\Delta L \pm 5 \mu\text{m} \rightarrow$ phase error $\Delta\phi \approx \pm 0.035\pi$.

Index drift $\Delta n \pm 0.01 \rightarrow$ uniform phase drift.

Coupler ratio error $\pm 2\%$.

Loss $\leq 0.2 \text{ dB/cm} \rightarrow$ notch shallowing.

Thermal crosstalk $-20 \text{ dB} \rightarrow$ phase cross-couple.

Laser λ drift $\pm 0.1 \text{ nm} \rightarrow$ phase drift.

6. Provisional Patent Claim Map

Ind. Claim 1: Guided-wave device with paths of length $\propto \log p$ and couplers/phasers implementing $\sum_{\mathbf{n}} (p^{\mathbf{n}} T_{\mathbf{n}} + p^{\mathbf{n}}(1-\mathbf{n}) T_{\mathbf{n}})$

Ind. Claim 2: Global phase control $\phi_{\mathbf{n}} = K \cdot \log p$ to scan spectral response.

Dep. Claims: SiN/SiO₂ platform at 1550 nm; adjoint symmetry measurement; finite-P truncation; acoustic waveguide extension

Trade-secret: heater calibration matrix, loss-equalization routine, routing compiler, optimal α, P, N scaling law.







